

Article

A Quantum-Inspired Ant Colony Optimization Algorithm for Parking Lot Rental to Shared E-Scooter Services

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Abstract: Electric scooter sharing mobility services have recently spread in major cities all around the world. However, the bad parking behavior of users has become a major source of issues, provoking accidents and compromising urban decorum of public areas. Reducing wild parking habits can be pursued by setting reserved parking spaces. In this work, we consider the problem faced by a municipality that hosts e-scooter sharing services and must choose which locations in its territory may be rented as reserved parking lots to sharing companies, with the aim of maximizing a return on renting and while taking into account spatial consideration and parking needs of local residents. Since this problem may result difficult to solve even for a state-of-the-art optimization software, we propose a hybrid metaheuristic solution algorithm combining a quantum-inspired ant colony optimization algorithm with an exact large neighborhood search. Results of computational tests considering realistic instances referring to the Italian capital city of Rome show the superior performance of the proposed hybrid metaheuristic.

Keywords: shared mobility; electric scooters; parking lot rental; combinatorial optimization; matheuristics; quantum computing; ant colony optimization

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1. Introduction

In recent times, shared mobility has spread and gained a lot of attention all around the world. Following the definition of a major organism like the United States Department for Transportation, it can be defined as a form of mobility that allows people to move through the shared use of vehicles (e.g., cars and bikes) on an as-needed basis and over short time periods [1], implementing the paradigm of Mobility-as-a-Service (MaaS) and pursuing a more sustainable way of moving in cities [2,3]. A major advantage of shared mobility is that it contributes to reducing the spread of car ownership, in turn contributing to decreasing traffic congestion, air pollution and CO₂ emissions, and thus benefitting all the urban community and not just individuals [4–6].

During the last decade, besides more traditional sharing services like car-sharing and bike-sharing, which, as the name clearly suggests, put cars and bicycles at the disposal of users, electric scooters (e-scooters) sharing services have experienced huge diffusion and development and have been launched in many cities throughout the globe. The spread of e-scooters, both in sharing and privately owned, has been favored by the fact that these vehicles are cheap to buy and maintain, easy to ride and park, and are very agile in city traffic due to their tiny size. Such features have been so appreciated that some recent studies have pointed out how e-scooter riding has substantially replaced biking and walking and reduced the use of cars in some cities [7–9]. For an exhaustive introduction to e-scooter mobility, we refer the reader to [10,11].

It has soon become apparent that the straightforwardness of riding and parking of e-scooters is a major advantage but also constitutes a major source of risks and issues: riders

have shown a tendency to hazardous driving behaviours and wild parking habits, which have led to an increase in accidents and decrease in urban decorum, as highlighted in many recent studies (e.g., [12–14]). As a consequence, municipalities have taken countermeasures, inflicting fines on e-scooter sharing companies and even banning them, e.g., in the French capital city of Paris [15,16].

For tackling such problems, a number of strategies have been proposed, ranging from more effectively controlling the flows of e-scooters through optimized geofencing (e.g., [17]), to better designing e-scooter regulations (e.g., [18,19]), to deploying staff for specifically correcting the wild parking of e-scooters and pursuing urban decorum (so-called “beautificators”, see [20]), even with the support of drones [21].

Another major strategy, on which we focus in this work, is represented by acting on parking availabilities and designing parking lots that are specifically reserved for e-scooters. This topic is not new in the literature, but we note that it has received limited attention for the e-scooter case. In [22], the Authors propose a fuzzy-based multi-criteria decision framework for the identification of e-scooter parking locations in urban areas, taking into account twelve criteria and evaluating three possible alternatives (i.e., free-floating operations, docking-station-based operations and hybrid operations, including geo-fences hubs located in catchment regions of public transportation). In [23], a data-driven approach is instead proposed, aimed at establishing the position of a given number of e-scooter parking facilities through a clustering algorithm that attempts to maximize a measure of trip capture. Also, ref. [24] proposes a data-driven approach for the identification of city areas characterized by high e-scooter parking demand and road segments with high e-scooter traffic, on which interventions by local authorities should be focused, improving infrastructures and concentrating the enforcement of road rules.

However, we note that there is still a gap to fill concerning the adoption of an Integer Programming-based approach for optimal e-scooter parking location and design. To this end, leveraging on our experience gained for the optimal design of car-sharing reserved parking in [25,26], we propose a new mathematical optimization model and algorithm that takes into account the specific features of e-scooter parking design, particularly taking into account the question of subtracting parking places to the cars of local residents.

In this paper, our main original contributions are:

1. characterizing the optimization problem faced by a municipality that hosts e-scooter sharing services and must choose which locations in its territory may be rented as reserved parking lots to sharing companies, with the aim of maximizing a return on renting, while taking into account spatial considerations and the parking needs of local residents;
2. modeling the problem by Mathematical Programming techniques, in particular, deriving a Binary Linear Program, which is centered around the use of binary decision variables for representing whether a potential parking lot is offered for rental or not;
3. proposing a modified Ant Colony Optimization solution algorithm that integrates techniques from Quantum Computing and Exact Neighborhood Search, since the optimization problem is NP-Hard and may prove challenging to solve even with state-of-the-art optimization software;
4. reporting and discussing results of computational tests showing the higher performance of our proposed metaheuristic solution approach with respect to a state-of-the-art solver.

The remainder of the paper is organized as follows. In Section 2, we characterize the parking lot rental problem and present a Binary Linear Programming problem for modeling it. In Section 3, we present the hybrid metaheuristic solution algorithm inspired by Quantum-based Ant Colony Optimization. In Section 4, we present and analyze the results of computational tests, whereas in Section 5, we conclude the paper by discussing directions for future work.

2. Problem Definition

We focus on the problem that a municipality deals with when it must choose the locations of lots to be rented for the parking shared e-scooters inside its administrated territory. Commonly, such territory is partitioned into a set of districts D and, in turn, every district $d \in D$ is partitioned into a set of subdistricts $S(d)$ (see Figure 1 for a real-world example). We also introduce the set of all subdistricts S , defined as $S = \cup_{d \in D} S(d)$ and we remark that any subdistrict belongs to one single district. Within a subdistrict s belonging to a district d , the municipality identifies the locations of a set of potential parking lots $L(d, s)$, among which the activated and deployed ones are chosen. Each lot $\ell \in L(d, s)$ is characterized by the number $n_{ds\ell}$ of e-scooters that it may host. Such a number depends on the size of the lot and we remark that each lot must be rented as a whole (no fraction of places can be rented). Each lot is also associated with a value $v_{ds\ell}$ that reflects the economic benefits of making it available to sharing services.

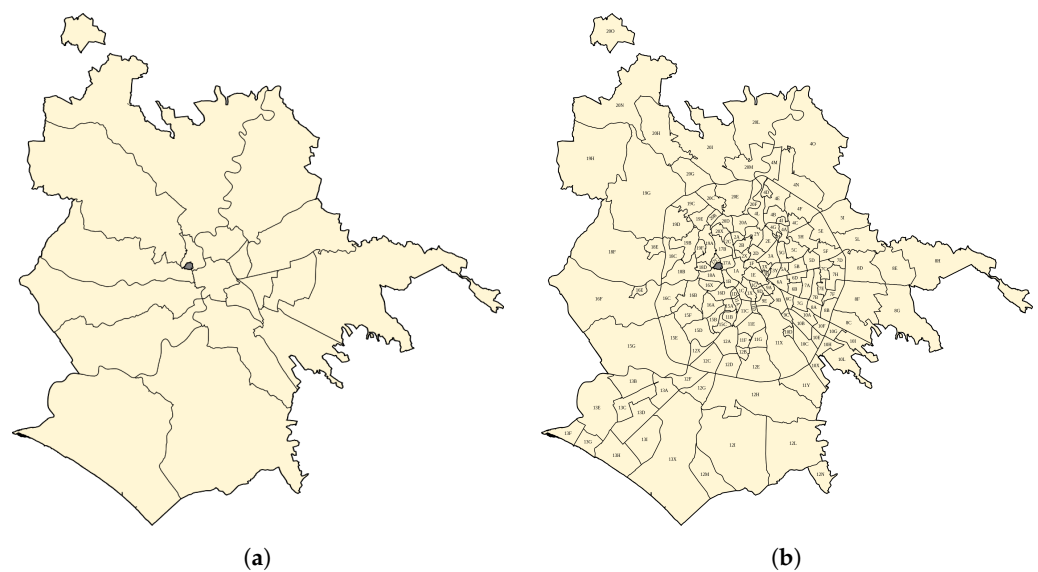


Figure 1. Visualization of the administrative division of the territory of Rome (“Municipi” and “Zone Urbanistiche”). (a) Districts of Rome, (b) Subdistricts of Rome.

When choosing which lots to rent, a municipality typically takes into account a lower bound LB_{ds}^{TOT} and an upper bound UB_{ds}^{TOT} on the total number of parking places that are made available to the sharing services in each subdistrict s of a district d . This takes into account the size of the fleets of sharing companies (for example, in the case of Rome, Italy, at most 3000 shared e-scooters are allowed to be deployed in the entire city by each of the three companies that operate in the city [27]).

Furthermore, an important aspect to take into account when modeling the rental problem is that not all the candidate lots identified by the municipality may be rented at the same time: these reserved lots “steal” space from other parking lots, in particular, those reserved for cars, and this may lead to discontent among local residents. We indeed consider the case in which the lots are positioned directly on the street, in order to avoid e-scooters riding on sidewalks for parking (riding on sidewalks constitutes a relevant source of risks for pedestrians). The discontent of residents is especially expected in cities congested with cars, where finding an empty parking lot may be problematic and cruising time for parking search tends to be high (this is the case of the Italian city of Rome, which we consider as a test case in our computational study). For modeling this, we first introduce a value $c_{ds\ell}$ that indicates the number of car parking places that are occupied by a lot ℓ and we then introduce an upper bound UB_{ds}^c denoting the highest number of car places that can be subtracted by e-scooter parking lots in each subdistrict s of a district d . This limit could and should be defined in concertation with local residents and, for example,

could be defined as a (very) small fraction of the total number of car places available in the subdistrict.

Finally, we consider the possibility for the municipality to classify all the potential lots according to classes: for instance, the class of a lot ℓ could be established by taking into account the number $n_{ds\ell}$ of places that ℓ hosts; as an alternative, as we have discussed with professionals of shared micromobility, the municipality could classify lots according to the nature of its location (e.g., residential or business) and could aim at having a fair balance between larger and smaller lots in its districts. If we denote by Γ the set of lot classes and by $\gamma(\ell) \in \Gamma$ the class of a lot ℓ , we introduce coefficients LB_d^γ, UB_d^γ to, respectively, represent the lower and upper bound on the number of lots of class $\gamma \in \Gamma$ that can be deployed in each district $d \in D$.

After having introduced all the required system elements, we can state the problem that we intend to model.

Definition 1 (The E-Scooter Parking Lot Rental Problem-ESPLR). *Given the set of districts D , each partitioned into a set of subdistricts $S(d)$, the set of parking lots that can be potentially rented $L(d, s)$, the bounds on the number of lots, car places subtracted and lot types $LB_{ds}^{TOT}, UB_{ds}^{TOT}, UB_{ds}^c, LB_d^\gamma, UB_d^\gamma$, the numbers $c_{ds\ell}$ of car places occupied by lots, the number of spaces $n_{ds\ell}$ hosted by lots and the values $v_{ds\ell}$ of the lots, the E-Scooter Parking Lot Rental Problem (ESPLR) is that of choosing the parking lots that are rented in each subdistrict, respecting all the bounds, with the aim of maximizing the total value of the rented lots.*

The ESPLR problem can be modeled as a Binary Linear Programming problem, including the decision variables, feasibility constraints and objective function that follow. The first component of the problem is represented by binary decision variables $x_{ds\ell} \in \{0, 1\}$, for each district $d \in D$, subdistrict $s \in S(d)$ and lot $\ell \in L(d, s)$, which assume value according to the following rules:

$$x_{ds\ell} = \begin{cases} 1 & \text{if lot } \ell \text{ in subdistrict } s \text{ of district } d \text{ is rented} \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

Then, we introduce a set of constraints for expressing the bounds on the total number of parking places that can be rented in each subdistrict:

$$LB_{ds}^{TOT} \leq \sum_{\ell \in L(d,s)} n_{ds\ell} \cdot x_{ds\ell} \leq UB_{ds}^{TOT} \quad \forall d \in D, s \in S(d) \quad (2)$$

Note, that in the previous constraints, a decision variable $x_{ds\ell}$ is multiplied by the corresponding number of parking places $n_{ds\ell}$ that the lot ℓ includes.

Then, we introduce a set of constraints for expressing the upper bound on the total number of car places that can be subtracted by e-scooter parking lots in each subdistrict:

$$\sum_{\ell \in L(d,s)} c_{ds\ell} \cdot x_{ds\ell} \leq UB_{ds}^c \quad \forall d \in D, s \in S(d) \quad (3)$$

Note that, in the previous constraints, the number of subtracted places is obtained by the multiplication of a decision variable $x_{ds\ell}$ with the coefficient $c_{ds\ell}$ expressing the number of car places occupied by the e-scooter parking lot ℓ .

Eventually, we introduce constraints for modeling the lower and upper bounds on the numbers of lots of class $\gamma \in \Gamma$ that can be deployed in each district:

$$LB_d^\gamma \leq \sum_{s \in S(d)} \sum_{\ell \in L(d,s): \gamma(\ell)=\gamma} x_{ds\ell} \leq UB_d^\gamma \quad \forall d \in D, \gamma \in \Gamma \quad (4)$$

We remark that, here, the second summation has the aim of selecting only those decision variables associated with lots of district d whose class coincides with the class γ of the constraint.

The objective function provides for maximizing the total value of rented lots, considering all districts and subdistricts:

$$\max \sum_{d \in D} \sum_{s \in S(d)} \sum_{\ell \in L(d,s)} v_{ds\ell} \cdot x_{ds\ell} \quad (5)$$

The overall optimization model, obtained by combining all the variables, constraints and objectives (1)–(5) is written as:

$$\begin{aligned} \max \sum_{d \in D} \sum_{s \in S(d)} \sum_{\ell \in L(d,s)} v_{ds\ell} \cdot x_{ds\ell} & \quad (\text{BLP-ESPLR}) \\ \text{LB}_{ds}^{\text{TOT}} \leq \sum_{\ell \in L(d,s)} n_{ds\ell} \cdot x_{ds\ell} \leq \text{UB}_{ds}^{\text{TOT}} & \quad \forall d \in D, s \in S(d) \\ \sum_{\ell \in L(d,s)} c_{ds\ell} \cdot x_{ds\ell} \leq \text{UB}_{ds}^c & \quad \forall d \in D, s \in S(d) \\ \text{LB}_d^\gamma \leq \sum_{s \in S(d)} \sum_{\ell \in L(d,s): \gamma(\ell)=\gamma} x_{ds\ell} \leq \text{UB}_d^\gamma & \quad \forall d \in D, \gamma \in \Gamma \\ x_{ds\ell} \in \{0, 1\} & \quad \forall d \in D, s \in S(d), \ell \in L(d,s) \end{aligned}$$

We denote this model by BLP-ESPLR and we can note that it is NP-Hard since it constitutes a variant of a generalization of the knapsack problem with minimum filling constraints (see [26]), introduced and proved to be NP-Hard in [28].

3. Quantum-Based ACO Algorithm

The model BLP-ESPLR is a Binary Linear Programming problem and, theoretically, could be solved by adopting any of the optimization solvers that are available commercially (e.g., IBM CPLEX [29]) or freely (e.g., GLPK [30]). However, even a powerful software like CPLEX, which may offer tremendously improved performance with respect to any free software, may encounter difficulties in solving fast and optimally real-world instances of BLP-ESPLR, which we recall to be an NP-Hard problem.

For solving BLP-ESPLR, we then propose to adopt a metaheuristic based on Ant Colony Optimization (ACO), which represents one of the most successful and widely-used nature-inspired metaheuristics [31,32]. Specifically, ACO belongs to the family of stochastic optimization algorithms, which employs randomness for exploring the solution space and includes other algorithms like the widely-known Particle Swarm Optimization (e.g., [33–35]) and colony search [36]). ACO draws inspiration from the behavior of ants searching for food and was proposed as a cooperative learning approach for solving hard combinatorial optimization problems [37]. Over the years, it has attracted a lot of attention and has been the subject of a huge number of theoretical and applied investigations (see, e.g., [38–42]).

In our case, we adopt a Quantum-inspired ACO (denoted here by QACO) algorithm proposed in [43], which improves the convergence of a canonical ACO algorithm by including qubit calculations from quantum computing for a generalization of the Travelling Salesman Problem including sustainability aspects. We refer the reader to [43] for exhaustive explanations of the algorithm. Here, we presume that the main feature of QACO is to refine the computational behavior of pheromone quantities used by ants for constructing feasible solutions to the problem, both in the initialization and update of pheromone values. The rationale at the basis of using and adapting QACO to our specific problem is that this algorithm provides an effective and efficient way to initialize the pheromone trails, whose setting is known to constitute a tricky task in general ACO (see, e.g., [44–46]), by using qubits values that are computed referring to the objective function of the optimization problem. With respect to QACO, we also propose to add an improvement phase at the

end of the solution construction phase. Specifically, we propose to integrate an *exact large neighborhood search* based on a *relaxation-based variable fixing* strategy, which attempts to further improve the best solution found in the QANT construction phase, following principles illustrated in [47].

The general structure of a canonical ACO algorithm is presented in Algorithm 1 and is fundamentally centered around a cycle where a number of feasible solutions are iteratively constructed, according to pheromone quantities that are cyclically updated in order to learn from the quality of solutions that have been previously built.

Algorithm 1 General canonical ACO Algorithm

```

1: while an arrest condition is not satisfied do
2:   ant-based solution construction
3:   pheromone trail update
4: end while
5: local search

```

At the end of the ant-based solution construction, we highlight the custom of running some kind of local search procedure in order to improve the best solution found (see e.g., [48]).

We specify below how the various steps and phases of the algorithm are specifically set up in our algorithmic approach.

3.1. Ant-Based Solution Construction

In any ACO algorithm, the phase of construction of solutions for the optimization problem is centered around a cycle executed by a set of agents (corresponding to ants) in which every agent/ant iteratively builds a solution moving from state to state. Each state corresponds to a partial solution of the optimization problem and an ant can make a move to another state to reach a complete solution by choosing moves from an available set. Talking more in mathematical optimization terms, a solution is said to be partial only if a subset of the decision variables has been assigned a value; instead, a solution is said to be complete if all the variables have been assigned a value. When an agent/ant executes a move, one or more decision variables are assigned a value and thus a partial solution becomes “more complete”. The selection of a move happens probabilistically: each move available to an ant in a state is associated with a probability computed on the basis of the attractiveness of choosing that move. Such attractiveness is called the pheromone measure and resembles the pheromone released by ants for marking promising food foraging paths: this pheromone measure is updated at the end of each construction phase by incrementing it for moves that have brought solutions of high quality and by decrementing it for moves that have brought solutions of bad quality.

To illustrate the algorithmic principles informally presented above more rigorously, we formally proceed to define the concept of state for the problem BLP-ESPLR. To conduct this, we must first remark that BLP-ESPLR is a purely binary linear program and consequently, constructing a complete solution requires an ant to fix the value of each decision variable $x_{ds\ell}$ to either 0 or 1. Then, we can define a Parking Lot State as follows:

Definition 2 (Parking Lot State–PLS). *Let $D \times S(d) \times L(d, s)$ be the set of triples (d, s, ℓ) that represent the rental of a lot ℓ in subdistrict s of district d . A Parking Lot State (PLS) identifies all the lots that are rented (thus, including triples corresponding to the decision variables $x_{ds\ell}$ with the value set to 1). Given a PLS, we can easily compute the corresponding number of e-scooter parking places and lots that are rented and the number of car parking places that are subtracted, thus having the possibility of easily verifying whether the constraints (2)–(4) are satisfied. We say that a PLS is partial when it does not satisfy the lower bounds $LB_d^\gamma, LB_{ds}^{\text{TOT}}$ of constraints (2), (4) and is instead complete when it satisfies them.*

If an ant is in a partial PLS, it can execute a move with the aim of reaching a (more) complete PLS. The move corresponds to renting one additional parking lot and consists of setting the value of a binary variable $x_{ds\ell}$ to 1, whose value has not yet been set. If we indicate by T the set of all triples (d, s, ℓ) representing all lots in subdistricts of districts and by T^{PLS} the triples rented in the state PLS, an ant in state PLS may execute a move choosing a triple $t = (d, s, \ell) \in T \setminus T^{\text{PLS}}$ with probability p_t set as:

$$p_t = \frac{\phi_t}{\sum_{\tau \in T} \phi_\tau} \quad \forall t \in T \setminus T^{\text{PLS}} \quad (6)$$

in which ϕ_t is the pheromone value of a move associated with triple $t \in T$. Executing the move of triple $t = (d, s, \ell)$ is equivalent to set $x_{ds\ell} = 1$.

3.2. Quantum-Based Pheromone Initialization and Update

The QACO algorithm by [43] is based on initializing the pheromone values $\phi_{t=(d,s,\ell)}$ referring to the theory of quantum computing (we direct the reader to the survey [49] for an exhaustive introduction to quantum technology and computation). Specifically, QACO employs a qubit (quantum bit—the basic unit of quantum information) with probabilities of states 0 and 1 represented by α^2 and β^2 , respectively, (thus it holds $\alpha^2 + \beta^2 = 1$).

As in [43], the value of α is computed for each possible move (in our case representing the rental of triple $t = (d, s, \ell)$) on the basis of the objective function coefficients and we set:

$$\alpha_{d,s,\ell} = \mu \cdot \frac{v_{ds\ell}}{\sum_{s \in S(d)} \sum_{\ell \in L(d,s)} v_{ds\ell}} \quad \forall t = (d, s, \ell) \in T \quad (7)$$

$$\beta_{d,s,\ell} = \sqrt{1 - \alpha_{d,s,\ell}^2} \quad \forall t = (d, s, \ell) \in T \quad (8)$$

in which $\mu = \text{rand}[0, 1]$ and $v_{ds\ell}$ is the value associated with renting lot ℓ .

The initial pheromone value $\phi_{d,s,\ell}$ of each triple (d, s, ℓ) is then set equal to $\alpha_{d,s,\ell}$, taking into account that the higher the value of α , the higher the attractiveness of the move since it is higher than the value $v_{ds\ell}$ of renting the lot (i.e., we initially set $\phi_{d,s,\ell} = \alpha_{d,s,\ell}$).

At the end of a cycle of solution construction of the set of ants, the pheromone values are updated in order to learn from the quality of solutions built and to model the natural fact that the ant pheromone fades away with time. We follow the approach proposed by [43] consisting of involving the quantum values in the update formula and we set:

$$\phi_{d,s,\ell}[i] = (1 - \epsilon) \cdot \phi_{d,s,\ell}[i - 1] + \frac{1}{(\beta_{ds\ell}^{\text{best}})^\lambda} \quad (9)$$

in which $\epsilon \in (0, 1)$ is the evaporation rate, $[i]$ and $[i - 1]$ are the index representing the cycle execution count, $\beta_{ds\ell}^{\text{best}}$ is the best value of $\beta_{ds\ell}$ found until iteration $[i]$ and λ is a parameter such that $\lambda \geq 2$.

3.3. MP-Based Improvement Search

Once the QACO-based solution construction and exploration have been completed, we use the best feasible solution found as a basis for finding other solutions associated with higher objective values. To this end, we run a large neighborhood search in an exact way. In this case, the adjective “exact” means that we formulate the search by means of Mathematical Programming (MP) techniques and we optimally solve the resulting optimization problem through state-of-the-art optimization software like CPLEX. The rationale for inspiring such an approach is that even though a powerful MP solver cannot optimally solve a large and complex problem, it may instead rapidly and efficiently solve suitable sub-problems. In our case, we identify and solve the sub-problems relying on the *Relaxation Induced Neighborhood Search* (RINS) proposed in [47]: given a feasible solution to an optimization problem, RINS searches for better solutions in a neighborhood defined

considering a linear relaxation of the problem. More formally and referring to our problem BLP-ESPLR, supposing that:

- $\bar{x}_{ds\ell}$ is a feasible solution of BLP-ESPLR (i.e., a binary solution satisfying all the constraints (2)–(4));
- $x_{ds\ell}^{LR}$ is the optimal solution of the linear relaxation of the problem (1)–(5) (i.e., a relaxed version BLP-ESPLR obtained by replacing the integrality requirement of each variable $x_{ds\ell} \in \{0, 1\}$ with its continuous relaxation $0 \leq x_{ds\ell} \leq 1$).

Then, for a chosen constant $\rho \in (0, 1)$, the neighborhood $\mathcal{N}(\bar{x})$ of the feasible solution $\bar{x}_{ds\ell}$ is defined as the set of feasible solutions of the sub-problem of BLP-ESPLR in which:

- if $\bar{x}_{ds\ell} = 0 \wedge x_{ds\ell}^{LR} \leq \rho$ then $x_{ds\ell} = 0$
- if $\bar{x}_{ds\ell} = 1 \wedge x_{ds\ell}^{LR} \geq 1 - \rho$ then $x_{ds\ell} = 1$
- else $x_{ds\ell} \in \{0, 1\}$.

In other words, a subset of the decision variables of the subproblem have the value fixed and such fixing happens if the value of a decision variable is sufficiently similar in both the feasible solution $\bar{x}$ and in the optimal solution x^{LR} of the linear relaxation. As discussed in [47], such similarity of values in the integral and relaxed problem provides a good indication that the variable is likely to have that value in feasible solutions of high quality.

More in detail, if we denote by I^1 and I^0 the set of indices of variables fixed to 1 and 0, respectively, according to the rules above, the subproblem for a feasible solution $\bar{x}$ can be written as:

$$\begin{aligned}
 & \max \sum_{d \in D} \sum_{s \in S(d)} \sum_{\ell \in L(d,s)} v_{ds\ell} \cdot x_{ds\ell} && \text{(BLP-ESPLR}(\bar{x})\text{)} \\
 & \text{LB}_{ds}^{\text{TOT}} \leq \sum_{\ell \in L(d,s)} n_{\ell} \cdot x_{ds\ell} \leq \text{UB}_{ds}^{\text{TOT}} && \forall d \in D, s \in S(d) \\
 & \sum_{\ell \in L(d,s)} c_{ds\ell} \cdot x_{ds\ell} \leq \text{UB}_{ds}^c && \forall d \in D, s \in S(d) \\
 & \text{LB}_d^{\gamma} \leq \sum_{s \in S(d)} \sum_{\ell \in L(d,s): \gamma(\ell)=\gamma} x_{ds\ell} \leq \text{UB}_d^{\gamma} && \forall d \in D, \gamma \in \Gamma \\
 & x_{ds\ell} = 1 && \forall (d, s, \ell) \in I^1 \quad (10) \\
 & x_{ds\ell} = 0 && \forall (d, s, \ell) \in I^0 \quad (11) \\
 & x_{ds\ell} \in \{0, 1\} && \forall d \in D, s \in S(d), \ell \in L(d, s)
 \end{aligned}$$

in which constraints (10) model the fixing of variables to 1 and (11) model the fixing of variables to 0.

We indicate this subproblem by BLP-ESPLR($\bar{x}$), pointing out that it is defined on the basis of an incumbent feasible solution $\bar{x}$.

Following the fixing of variables, the resulting sub-problem is solved by software like CPLEX with the aim of being optimally solved and hopefully finding a better solution. We note that an optimal solution to the subproblem clearly constitutes a feasible solution for the original complete problem. The exact search in the neighborhood is typically conducted within a time limit.

The overall pseudocode of our hybrid metaheuristic, denoted by HQ-ACO, is presented in Algorithm 2. The initialization requires the definition of the quantum values $\alpha_{ds\ell}$, $\beta_{ds\ell}$, which are employed to initialize the pheromone values $\phi_{d,s,\ell}$ associated with activating a lot ℓ in subdistrict s of district d . Then an outer loop with a time-limit arrest condition is executed. At every execution of the loop, M ants iteratively build M complete Parking Lot States, thus defining M feasible solutions to the optimization problem. If a solution is better than the best found, an update is conducted. At the end of each construction phase of the M ants, the pheromone trails are updated. Once the time limit of the outer loop is

reached, the quantum-ACO construction phase ends and the best solution found is used as a basis to define the neighborhood used in RINS.

Algorithm 2 Hybrid Quantum-based ACO (HQ-ACO)

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1: Let  $x^{\text{best}}$  denote the best solution found and set its corresponding value to
    $v(x^{\text{best}}) := -\infty$ 
2: Calculate the values  $\alpha_{ds\ell}, \beta_{ds\ell}$  by Formulas (7) and (8)
3: Initialize the pheromone values  $\phi_{d,s,\ell}$ 
4: while a global time limit is not reached do
5:   for  $m := 1$  to  $M$  do
6:     Build a complete Parking Lot State (PLS) and let  $x^{\text{PLS}}$  be the corresponding
       solution
7:     if  $v(x^{\text{PLS}}) \geq v(x^{\text{best}})$  then
8:       Update the best solution found by setting  $x^{\text{best}} := x^{\text{PLS}}$  and  $v(x^{\text{best}}) =:$ 
        $v(x^{\text{PLS}})$ 
9:     end if
10:  end for
11:  Update the pheromone values by Formula (9)
12: end while
13: Run the exact neighborhood search solving problem BLP-ESPLR( $x^{\text{best}}$ ) for improving
    $x^{\text{best}}$ 

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4. Computational Tests

We tested the performance of the optimization model BLP-ESPLR and of the hybrid metaheuristic HQ-ACO using realistic data related to the city of Rome, Italy. The considered data instances have been defined not only taking into account the regulations of the city (in particular, the more recent updates regulating e-scooter sharing services imposing, e.g., to reduce the number of allowed companies to three, extending the overall service area to outer districts, see [27,50,51]) but also on the basis of discussions with sharing mobility professionals of companies that have operated in Rome.

For defining the set of districts and subdistricts, we have considered the administrative division of the territory of Rome and, in each subdistrict, we have identified a set of parking potentially available for renting. The lots have been defined taking into account the relevance of the subdistricts, their position within Rome and their nature (business, residential, peripheral, presence of points of interest, etc.). The types of lots have been set to 6, taking into account their size and location with respect to landmarks and every lot may contain from 3 to 15 e-scooters. Larger lots are especially located in central areas and close to major public transportation hubs and are meant to host larger sets of e-scooters from all three active companies. The bounds on the total number of places, the bounds on the number of slot types and the upper bound on the number of car places that can be subtracted by reserved e-scooter lots have been defined by taking into account the features of the subdistricts (type of residents, public transportation presence, features of the urban fabric).

The algorithm HQ-ACO has been coded in C/C++ and executed on a 2.70 GHz Windows machine equipped with 8 GB of RAM and using IBM ILOG CPLEX 12.5 as optimization software. Both CPLEX and HQ-ACO ran with a time limit of 3600 s. In the case of IQ-ACO, 3000 s are reserved for the solution construction phase and the remaining 600 s are reserved for the RINS-based exact neighborhood search. The ACO construction phase employs 10 ants with an evaporation factor $\epsilon = 0.1$, and the RINS search uses $\rho = 0.1$.

For test purposes, we have considered 20 instances of increasing size from about 600 to about 4300 potential slots associated with up to about 33,000 potential single parking places. All ten instances consider a distribution of the parking lots that extends to all the districts of Rome, reflecting the enlargement of the overall service area imposed by the new e-scooter sharing plan agreed with the municipality of Rome. The distribution aims, in particular, at covering new outer zones that are located well beyond the railway ring

surrounding more central districts (in Italian, the so-called “Anello Ferroviario”) and that were excluded or poorly served in the previous regulation setting of the city of Rome. Lots located in peripheral areas are characterized by a (much) lower value than those in central historical districts. In the case of the first ten instances (ID from 1 to 10), given the lower total number of lots, the density of place distributions across the subdistricts is lower. In the case of instances with ID from 11 to 20, the number of potential lots sensibly increases, in particular, with the aim of potentially offering a higher density in central districts that are more afflicted by e-scooter wild parking. These lots in central areas tend to naturally have a substantially higher value than those located in suburbs. The upper bounds UB_{ds}^{TOT} and UB_{ds}^c have been set equal to 10% of the total number of potential places in each subdistrict and to 5% of the total number of car places in each subdistrict.

In Table 1, we visualize the results of the computational tests and we denote by “ID” the index number of the data instance, by “Lots” the total number of potential lots available, by “Places” the total number of parking places (we recall that each e-scooter parking lot may contain a different number of places). Then, in the set of columns that follow: (1) “Places (act)” is the number of places that are offered for rental in the best solution found; (2) “ Δv (best)” and “ $\Delta v\%$ (best)” are the difference expressed in EUR and the corresponding percentage difference between the value of the best solution found by CPLEX within the time limit and the value of the best solution found by HQ-ACO within the limit (we note that a positive percentage indicates a better performance of HQ-ACO with respect to CPLEX); (3) “ Δv (avg)” and “ $\Delta v\%$ (avg)” are the difference expressed in EUR and the corresponding percentage difference between the value of the best solution found by CPLEX within the time limit and the average value of feasible solutions found by HQ-ACO within the limit (also in this case, a positive percentage indicates a better performance of HQ-ACO with respect to CPLEX).

Looking at the results of Table 1, it is possible to make the following major observations.

- For the first three instances of smaller size 1–3, CPLEX and HQ-ACO find an optimal solution to the problem (in this case, the percentage difference of their value $\Delta v\%$ is then equal to zero). For these instances, since CPLEX finds an optimal solution, we also note that the average value of feasible solutions found by HQ-ACO is naturally smaller than the best solution found by CPLEX.
- For all the remaining instances, CPLEX is not able to find an optimal solution within the time limit and, for those of larger size from 13 to 20, it even encounters difficulties in identifying good quality solutions. This should not be a surprise: even though it has a neat mathematical structure, the problem BLP-ESPLR is NP-Hard and its difficulty rapidly increases as the size of the instances grows and the distribution of lots and parking becomes more articulated and introduces more complex correlation through the three distinct set of bound constraints. In contrast, HQ-ACO is able to find solutions of higher quality that, on average guarantee an increase in value of 13% and may exceed 20% in the case of the larger instances, obtaining more than a 25% increase for the instance with ID = 19. Looking more closely at the structure of the best solutions found, it is interesting to note that they tend to activate as many lots as possible including, in particular, those with a higher number of spaces, since they are typically associated with higher lot rental value per single space. Thanks to multiple bound constraints involving subdistricts and lot types, the distribution of the lots results fair, without neglecting outer districts that are often less considered when designing public transportation plans.
- For all the instances from ID 4 to ID 20, the average value of the feasible solutions identified by HQ-ACO is always higher than that of the best solution found by CPLEX. Naturally, this situation does not hold in the case of the three instances with ID 1–3 for which CPLEX finds an optimal solution. In the case of larger instances, the average value can be more than 10% higher with respect to CPLEX, granting a substantial amount of additional money from rental.

- Concerning computational time, looking at the final set of columns of the table, it can be noted that CPLEX needs a substantial amount of time to identify its best solution, taking more than 3000 s for most of the instances, with an average increase of about 28% with respect to the time that HQ-ACO requires to find its best solution.

In Figure 2, we visually compare the value of the best solution found by CPLEX and HQ-ACO and the average value of solutions found by HQ-ACO within the time limit. The figure clearly shows that both the best and average value of solutions found by HQ-ACO grants an increase with respect to CPLEX. Such higher performance in terms of objective function value is particularly evident for the larger instances I16–I20, in the case of which HQ-ACO grants an increase in value that is about 50k EUR on average for the best case and about 30k EUR on average for the average case.

The results reported in Table 1, Figure 2 and discussed above highlight the superior performance of the hybrid heuristic HQ-ACO with respect to a state-of-the-art solver, pointing out its ability to faster find solutions of higher quality, and thus shows the computational advantages of adopting it.

Table 1. Computational results.

ID	Lots	Places	Places (act)	Δv (best)	$\Delta v\%$ (best)	Δv (avg)	$\Delta v\%$ (avg)	t (HQ-ACO)	t (CPLEX)	$\Delta t\%$
1	610	2056	198	0	0	−644	−3.7	2109	2783	32.0
2	676	2529	247	0	0	−1365	−5.2	1922	2468	28.4
3	761	2765	275	0	0	−2740	−7.8	2175	2831	30.2
4	797	3387	324	2081	6.7	746	2.4	2484	3442	38.6
5	947	3260	335	2234	5.6	1397	3.5	2743	3276	19.4
6	995	3609	355	3649	7.8	2012	4.3	2956	3489	18.0
7	1167	4474	441	6638	12.3	2969	5.5	2508	3222	28.5
8	1160	4651	466	6008	9.8	2453	4.0	2497	3305	32.4
9	1262	5235	518	7932	13.6	3267	5.6	2561	3411	33.2
10	1459	4713	445	5862	12.4	2979	6.3	2465	3260	32.3
11	1750	7246	704	11831	16.8	6902	9.8	2344	2895	23.5
12	1895	8803	866	12846	13.1	4609	4.7	2423	3277	35.2
13	2071	9694	957	19448	17.2	8707	7.7	2734	3304	20.8
14	2282	9689	978	23227	22.3	9895	9.5	2475	3197	29.2
15	2736	11427	1166	22226	18.2	11968	9.8	3009	3462	15.1
16	3030	15960	1553	29710	16.9	17581	10.0	2429	3198	31.7
17	3257	17034	1673	41335	24.3	19222	11.3	2363	3261	38.0
18	3713	18214	1787	35454	19.4	24123	13.2	2711	3367	24.2
19	3782	27188	2687	68996	25.6	40158	14.9	2477	3450	39.3
20	4354	33252	3272	72891	20.6	47769	13.5	2874	3566	24.1

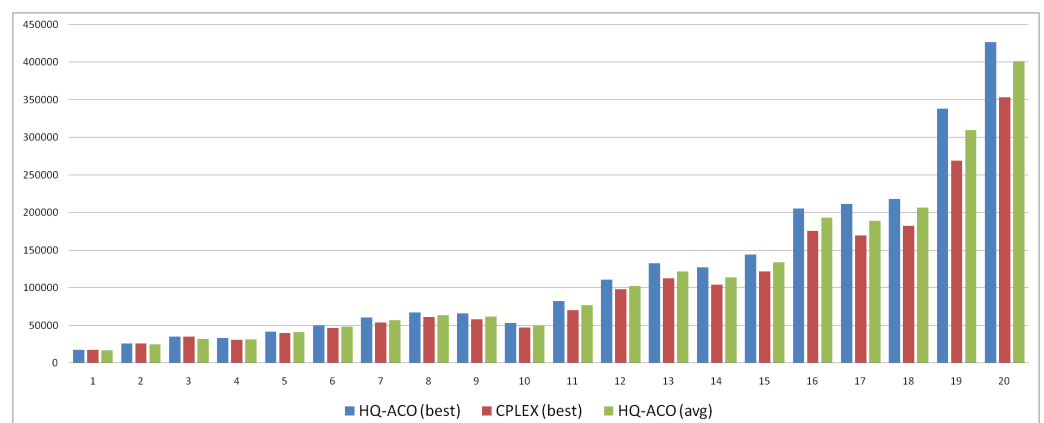


Figure 2. Value (in EUR) of the solutions found by CPLEX and HQ-ACO for the 20 instances

5. Conclusions and Future Work

We have investigated the development of an optimization approach for modeling and solving the problem faced by a municipality that hosts e-scooter sharing services and intends to rent reserved parking spaces to these services in order to reduce the issues associated with wild parking and promote higher driving safety. We proposed a Binary Linear Programming model to represent the model and, given its NP-hardness, we proposed a hybrid metaheuristic combining a quantum-inspired ant colony optimization algorithm with an exact large neighborhood search. Computational tests on realistic instances have shown the superior performance of the proposed solution algorithm with respect to state-of-the-art optimization software, providing a tool that could be adopted by municipalities for designing their parking space rental offers, and also easily allowing the modeling of the fair distribution of reserved parking over the territory. In future work, from an algorithmic point of view, we plan to enhance the performance of the solution algorithm, in particular, refining the critical pheromone function by better integrating valuable information that can be obtained from polyhedrally strong formulations of a mathematical programming model. From a more applied point of view, we intend instead to generalize the problem, possibly better integrating the uncertainty of user behavior through Robust Optimization.

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